Requirements Engineering

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Elaboration

- Analyze, model, specify…
- Some Analysis Techniques
  - Data Flow Diagrams (DFD)
  - Usecase Diagram
  - Object Models (ER Diagrams, Abstract class diagrams)
  - Decision Tables
  - State Diagrams (Statecharts)
  - Fence Diagrams
  - Event Tables
  - Petri Nets
  - Event Traces (Message Sequence Charts)

Flow-oriented, Scenario-based, Class-based, Behavioral

Decision Table

- It is a tabular representation of a functional specification that maps events and conditions to appropriate response or action
- The specification is formal because the inputs (events and conditions) and outputs (actions) may be expressed in natural language
- If there is *n* input conditions, there are 2^n possible combination of input conditions
- Combinations map to the same set of result can be combined into a single column

Example

Our university awards upto 100% scholarship to talented students using a University Scholarship Award System (USAS). The university also awards 25% fee waiver on kinship basis using the same system. To award the scholarships, the university maintains very high standards. Students having less than 90% marks in matriculation do not qualify for any scholarship and are given 0% fee waiver . Students scoring 90% or above in matriculation exam are assessed using this program on the basis of their performance in intermediate exam and admission test. Fee waiver is granted as per following rules:

- If a student scores 90% or above in intermediate and admission test, grant 100% fee waiver
- If a student scores less than 90% in only one of them, grant 50% fee waiver
- If a student scores less than 90% in both of them and has valid kinship, grant 25% fee waiver
- If a student scores less than 90% in both of them and does not have valid kinship, grant 0% fee waiver

Example

	Rule 1	Rule 2	Rule 3	Rule 4	Rule 5	Rule 6	Rule 7	Rule 8	Rule 9 -16
$Marks_{SSC}$ >= 90%	T	T	T	T	T	T	T	T	F
$Marks_{Inter}$ >= 90%	F	F	F	F	T	T	T	T	-
$Marks_{Test}$ >= 90%	F	F	T	T	F	F	T	T	-
Valid Kinship?	F	T	F	T	F	T	F	T	-
Grant 0% waiver	x								x
Grant 25% waiver		x							
Grant 50% waiver			x	x	x	x			
Grant 100% waiver							x	x	

Exercise

We need to develop a software system that, as a subtask, requires to determine if the values entered as input by the user will form a triangle or not. The software also determines the type of the triangle if the input values form a triangle.

The three input values A, B, and C are integer values and each corresponds to one side of a triangle. Based on the inputs, the system outputs the type of the triangle i.e. Equilateral, Isosceles, Scalene, NotATriangle. The NotATriangle output is provided if A, B, and C do not form a triangle. In order to form a triangle, the sides A, B, and C must meet the following constraints:

C1: $A < B + C$

C2: $B < A + C$

C3: $C < A + B$

The type of triangle is decided based on the following:

If all sides are equal then it is **Equilateral**

If exactly two sides are equal then it is **Isosceles**

If no pair of sides is equal then it is **Scalene**

If A, B, C do not satisfy any of the constraints C1, C2, C3 then it is **Not A Triangle**

Provide a complete decision table that models the information provided above. <u>Each condition must be expressed in terms of one or more input values (i.e. A, B, C) only. [Hint: Most of these conditions shall be simple (not compound).]</u>

Example

We need to develop a game scenario where there can be multiple buildings and roads. There can be different runnable objects on the roads having different colours, number of doors, and number of wheels. Each of the runnable objects can start, stop, and run in its own way. The runnable objects include bicycle, carts, vehicles (like rickshaw, car, van, truck etc.). Each road has a different width, length, curving angle and knows the buildings and their positions on its sides. The buildings know their own positions and have different number of windows and doors. These windows and doors have different dimensions (height, width) and opening mechanism (sliding, pushing). Doors are passable and windows are not. The runnable objects know the road they are running on. The runnable objects are driven by the players (or similar characters in the game) who can use the run, start, stop behaviours of the object when required.